Conceptual question · Unit 2 · 2 points · 15 min · Conceptual / fast

C03 — Energy of a pendulum hanging from a slider: one particle, the system, and a driven support

A slider A of mass M moves along a smooth, fixed horizontal rail (the x axis direction of the inertial frame $S_0 = \{O; \mathbf{i}, \mathbf{j}\}$, y upward, gravity $-g\,\mathbf{j}$); its abscissa is x_A. A bob P of mass m hangs from A on a massless, inextensible string of length ℓ that stays taut. The angle θ is measured from the downward vertical, positive towards $+x$, and $\mathbf{e}_r = \sin\theta\,\mathbf{i} - \cos\theta\,\mathbf{j}$ (from A to P), $\mathbf{e}_\theta = \cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j}$. The motion is planar. Decide whether each statement is TRUE or FALSE and justify the answer. A verdict without a justification scores nothing.

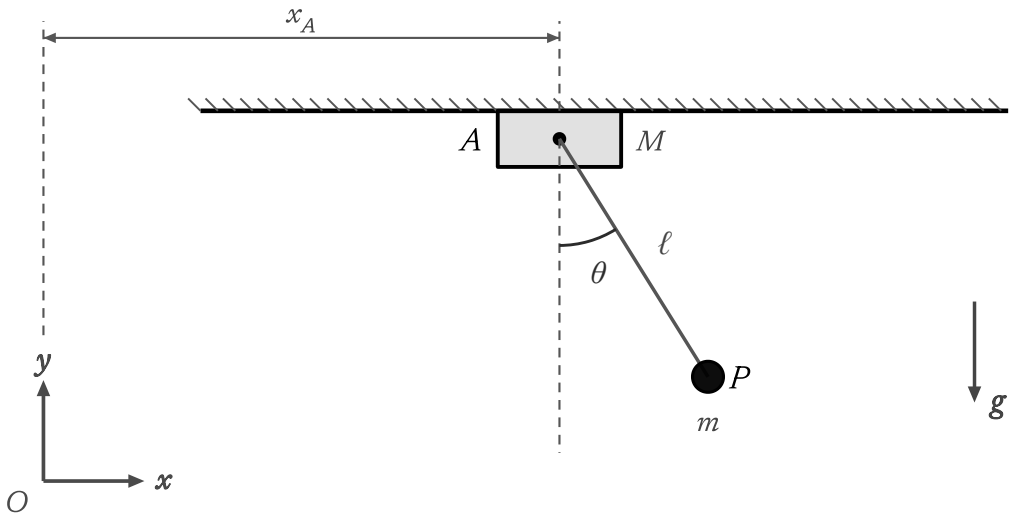


Figure 1. Slider A (mass M) on a smooth horizontal rail, abscissa x_A; bob P (mass m) on a string of length ℓ, angle θ from the downward vertical, positive towards $+x$.

Part (a) · 1 point

Free slider

*The slider moves freely on the rail. Statement: "The mechanical energy of the bob alone, $E_P = \frac{1}{2}m|\mathbf{v}_P|^2 + mg\,y_P$, is **not** conserved, but the mechanical energy of the system slider plus bob, $E = \frac{1}{2}M\dot{x}_A^2 + E_P$, is conserved."*

Step 1. Power of the tension on the bob

The string pulls P towards A: $\mathbf{T}_P = -T\,\mathbf{e}_r$, $T \geq 0$. The absolute velocity of P is $\mathbf{v}_P = \dot{x}_A\,\mathbf{i} + \ell\dot{\theta}\,\mathbf{e}_\theta$ (derivative of $\mathbf{r}_P = x_A\mathbf{i} + \ell\mathbf{e}_r$ with $\dot{\mathbf{e}}_r = \dot{\theta}\,\mathbf{e}_\theta$). With $\mathbf{e}_r \cdot \mathbf{i} = \sin\theta$ and $\mathbf{e}_r \cdot \mathbf{e}_\theta = 0$:

$$\dot{W}_{T,P} = \mathbf{T}_P \cdot \mathbf{v}_P = -T\dot{x}_A \sin\theta$$

The tension is orthogonal to the velocity of P **relative to** A, but not to its absolute velocity. By the work-energy theorem with the weight as the only conservative force on P (Notes §4.5), $\dot{E}_P = -T\dot{x}_A\sin\theta$, which is not zero whenever the slider moves with the string inclined: E_P is not conserved.

Step 2. The system

On A act $M\mathbf{g}$ and the rail reaction (both vertical, power zero against $\dot{x}_A\mathbf{i}$: smooth rail) and the string force $\mathbf{T}_A = +T\,\mathbf{e}_r$, of power $T\dot{x}_A\sin\theta$. Hence $\frac{d}{dt}\left(\tfrac{1}{2}M\dot{x}_A^2\right) = T\dot{x}_A\sin\theta$ and

$$\dot{E} = T\dot{x}_A\sin\theta - T\dot{x}_A\sin\theta = T\,\mathbf{e}_r\cdot(\mathbf{v}_A - \mathbf{v}_P) = -T\ell\dot{\theta}\,\mathbf{e}_r\cdot\mathbf{e}_\theta = 0$$

The internal pair of an inextensible string does no net work (Notes §4.4), so E is conserved. The statement is TRUE.

Step 3. How much energy moves from P to A

The horizontal external force is zero, so $p_x = (M+m)\dot{x}_A + m\ell\dot{\theta}\cos\theta$ is conserved, equal to 0 for a release from rest at θ_0. At $\theta = 0$: $\dot{x}_A = -\dfrac{m\ell\dot{\theta}}{M+m}$. At $\theta = 0$, $\mathbf{e}_\theta = \mathbf{i}$, so $\mathbf{v}_P = (\dot{x}_A + \ell\dot{\theta})\,\mathbf{i}$ with $\dot{x}_A + \ell\dot{\theta} = \dfrac{M\ell\dot{\theta}}{M+m}$, and $y_P = -\ell\cos\theta$. Energy conservation between the release (θ_0, at rest) and $\theta = 0$:

$$\frac{1}{2}M\frac{m^2\ell^2\dot{\theta}^2}{(M+m)^2} + \frac{1}{2}m\frac{M^2\ell^2\dot{\theta}^2}{(M+m)^2} - mg\ell = -mg\ell\cos\theta_0$$

The two kinetic terms add up to $\dfrac{Mm(m+M)\ell^2\dot{\theta}^2}{2(M+m)^2} = \dfrac{Mm\,\ell^2\dot{\theta}^2}{2(M+m)}$, so $\dfrac{Mm\,\ell^2\dot{\theta}^2}{2(M+m)} = mg\ell(1-\cos\theta_0)$, which gives $\ell\dot{\theta}^2 = \dfrac{2g(1-\cos\theta_0)(M+m)}{M}$ and

$$\tfrac{1}{2}M\dot{x}_A^2 = \frac{M}{2}\,\frac{m^2\ell^2\dot{\theta}^2}{(M+m)^2} = \frac{m^2 g\ell\,(1-\cos\theta_0)}{M+m} > 0$$

which is exactly what E_P has lost.

KEY POINTS

- TRUE.
- Tension on P: $-T\mathbf{e}_r$, power $-T\mathbf{e}_r \cdot \mathbf{v}_P = -T\dot{x}_A \sin\theta \neq 0$ in general, so $\dot{E}_P = -T\dot{x}_A \sin\theta$.
- Tension on A: $+T\mathbf{e}_r$, power $+T\dot{x}_A \sin\theta$; the pair has total power $T\mathbf{e}_r \cdot (\mathbf{v}_A - \mathbf{v}_P) = 0$ (inextensible string); rail reaction and $M\mathbf{g}$ are vertical, orthogonal to $\dot{x}_A\mathbf{i}$: $\dot{E} = 0$.
- Example: released from rest at θ_0, at $\theta = 0$ the slider has $\frac{1}{2}M\dot{x}_A^2 = \dfrac{m^2 g\ell(1-\cos\theta_0)}{M+m}$, energy that P has lost.

Part (b) · 1 point

Driven slider

A motor now imposes $x_A(t) = b\sin(\Omega t)$. Let S_1 be the frame that translates with A. Statement: "In S_1 the tension does no work on P and the weight is conservative; therefore $E_{rel} = \frac{1}{2}m\ell^2\dot{\theta}^2 - mg\ell\cos\theta$ is a first integral of the motion of the bob."

Step 1. Forces and work in S_1

S_1 translates with acceleration $\ddot{x}_A\mathbf{i}$, so (Notes §4.9) the forces on P in S_1 are $m\mathbf{g}$, $-T\mathbf{e}_r$ and the inertia force $\mathbf{F}_{in} = -m\ddot{x}_A\,\mathbf{i}$ (no Coriolis or centrifugal terms: $\boldsymbol{\omega}_{10} = \mathbf{0}$). With $\mathbf{v}_{rel} = \ell\dot{\theta}\,\mathbf{e}_\theta$ the tension does no work, and the weight has the potential $-mg\ell\cos\theta$. The power of the inertia force is

$$\mathbf{F}_{in}\cdot\mathbf{v}_{rel} = -m\ddot{x}_A\,\ell\dot{\theta}\,(\mathbf{i}\cdot\mathbf{e}_\theta) = -m\ddot{x}_A\,\ell\dot{\theta}\cos\theta$$

Step 2. Rate of change of E_{rel}

Projecting $m\mathbf{a}_{rel} = m\mathbf{g} - T\mathbf{e}_r + \mathbf{F}_{in}$ on $\mathbf{e}_\theta$: $m\ell\ddot{\theta} = -mg\sin\theta - m\ddot{x}_A\cos\theta$. Differentiating E_{rel} with the chain rule, $\frac{d}{dt}\cos\theta = -\sin\theta\,\dot{\theta}$:

$$\dot{E}_{rel} = m\ell\dot{\theta}\left(\ell\ddot{\theta} + g\sin\theta\right) = -m\ddot{x}_A\,\ell\dot{\theta}\cos\theta$$

With $\ddot{x}_A = -b\Omega^2\sin(\Omega t)$: $\dot{E}_{rel} = mb\Omega^2\ell\,\sin(\Omega t)\,\dot{\theta}\cos\theta$. For example at $\Omega t = \pi/2$ with $\theta = 0$ and $\dot{\theta} \neq 0$ it equals $mb\Omega^2\ell\dot{\theta} \neq 0$. E_{rel} is not a first integral: the statement is FALSE.

Step 3. Why no other energy works either

For a constant $\ddot{x}_A = a_0$ (as in a uniformly accelerating support) the inertia force would be uniform and conservative, with potential $ma_0\ell\sin\theta$. Here, if we try $E^* = E_{rel} + m\ddot{x}_A(t)\,\ell\sin\theta$, the explicit time dependence leaves

$$\dot{E}^* = m\,\frac{d^3x_A}{dt^3}\,\ell\sin\theta = -mb\Omega^3\cos(\Omega t)\,\ell\sin\theta \neq 0$$

The motor keeps pumping energy in and out of the bob.

KEY POINTS

- FALSE.
- In S_1: $\mathbf{v}_{rel} = \ell\dot{\theta}\,\mathbf{e}_\theta$, $-T\mathbf{e}_r \cdot \mathbf{v}_{rel} = 0$ (this part is true).
- Equation: $\ell\ddot{\theta} = -g\sin\theta - \ddot{x}_A\cos\theta$; $\dot{E}_{rel} = -m\ddot{x}_A\,\ell\dot{\theta}\cos\theta = mb\Omega^2\ell\sin(\Omega t)\,\dot{\theta}\cos\theta \neq 0$.
- The inertia force depends explicitly on t: even $E_{rel} + m\ddot{x}_A\ell\sin\theta$ changes, at the rate $m\,\dfrac{d^3x_A}{dt^3}\,\ell\sin\theta$.

Interpretation and checks

- **Dimensional check:** $T\dot{x}_A$ is N m s^{-1} = W; $mb\Omega^2\ell\dot{\theta}$ is kg m s^{-2} m s^{-1} = W; $m^2g\ell/(M+m)$ is J.
- **Limits:** $M \to \infty$ makes the slider a fixed support ($\dot{x}_A \to 0$): the energy transfer $m^2g\ell(1-\cos\theta_0)/(M+m) \to 0$ and E_P is conserved. $b \to 0$ in (b) gives the ordinary pendulum and $\dot{E}_{rel} \to 0$.
- **Common mistakes in the exam:** (1) "the tension is perpendicular to the velocity, so it does no work": true only for the velocity relative to the point the string is tied to; (2) forgetting the inertia force when writing the energy in a non-inertial frame; (3) treating a time-dependent inertia force as conservative.